

Class 9 biology – The Fundamental Unit of Life (NCERT Solutions)

Q1. Difference between plant cell and animal cell

Answer:

Feature	Plant Cell	Animal Cell
Cell wall	Present	Absent
Shape	Fixed (rectangular)	Irregular/round
Vacuole	Large central vacuole	Small or absent
Plastids	Present (chloroplast)	Absent
Mode of nutrition	Autotrophic	Heterotrophic

Q2. Difference between prokaryotic and eukaryotic cells

Answer:

Feature	Prokaryotic Cell	Eukaryotic Cell
Nucleus	Absent	Present
Size	Small	Large
Organelles	No membrane-bound organelles	Present
DNA	Not well defined	Well defined

Q3. What if plasma membrane breaks?

Answer:

Cell contents will leak out, leading to **cell death**.

Q4. If Golgi apparatus is absent

Answer:

- No packaging and transport of materials
 - Proteins and lipids cannot be properly processed
 - Cell functions will stop
-

Q5. Powerhouse of the cell

Answer:

Mitochondria → produces energy (ATP) through respiration.

Q6. Where are lipids and proteins synthesized?

Answer:

- Lipids → Smooth Endoplasmic Reticulum (SER)
 - Proteins → Ribosomes
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Q7. How does Amoeba obtain food?

Answer:

By **endocytosis** using pseudopodia (engulfing food).

Q8. What is osmosis?

Answer:

Movement of water from **higher concentration to lower concentration through a semi-permeable membrane**.

Q9. Potato experiment

Answer:

(i) Water gathers in B and C due to **osmosis** (higher solute concentration inside).

(ii) Potato A is a **control experiment**.

(iii)

- In A → no solute, so no osmosis

- In D → boiling destroys membranes, so osmosis does not occur
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Q10. Types of cell division

Answer:

- Growth and repair → **Mitosis**
 - Formation of gametes → **Meiosis**
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Extra Questions

Q11. Organelles with genetic material

Answer:

- Mitochondria
 - Plastids
-

Q12. If cell organization is destroyed

Answer:

Cell will **die** because all functions stop.

Q13. Why lysosomes are suicide bags?

Answer:

They contain digestive enzymes that can destroy the cell.

Q14. Where are proteins synthesized?

Answer:

At **ribosomes**.

Q15. Movement of substances in cell

Answer:

- $\text{CO}_2 \rightarrow$ Diffusion
 - Water $\rightarrow$ Osmosis
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Q16. Why plasma membrane is selectively permeable?

Answer:

It allows **some substances to pass while restricting others.**

Q17. Who discovered cells?

Answer:

Robert Hooke (1665).

Q18. Why cell is structural and functional unit?

Answer:

- All organisms are made of cells
 - All life processes occur inside cells
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Key Points for Revision

- Cell = basic unit of life
 - Mitochondria = energy
 - Ribosomes = protein synthesis
 - Osmosis = water movement
 - Prokaryotes vs Eukaryotes $\rightarrow$ important
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Class 9 biology – Tissues (NCERT Solutions)

Q1. Define the term “tissue”.

Answer:

A tissue is a group of similar cells performing a specific function.

Q2. Elements of xylem

Answer:

Xylem consists of **4 elements**:

- Tracheids
 - Vessels
 - Xylem fibres
 - Xylem parenchyma
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Q3. Simple vs Complex tissues

Answer:

Simple Tissue	Complex Tissue
Made of one type of cells	Made of different types of cells
Example: Parenchyma	Example: Xylem, Phloem
Perform similar function	Perform different functions

Q4. Difference based on cell wall

Answer:

Tissue	Cell Wall
Parenchyma	Thin
Collenchyma	Thick at corners

Tissue	Cell Wall
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Sclerenchyma	Very thick and lignified
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Q5. Functions of stomata

Answer:

- Exchange of gases (CO_2 , O_2)
 - Loss of water through transpiration
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Q6. Types of muscle fibres (diagram)

Answer (theory for exam):

- Striated $\rightarrow$ long, cylindrical, striations present
 - Smooth $\rightarrow$ spindle-shaped, no striations
 - Cardiac $\rightarrow$ branched, striated
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Q7. Function of cardiac muscle

Answer:

Helps in **continuous rhythmic contraction of the heart** to pump blood.

Q8. Differences between muscles

Answer:

Feature	Striated	Unstriated	Cardiac
Structure	Striped	No stripes	Striped + branched
Control	Voluntary	Involuntary	Involuntary
Location	Skeleton muscles	Internal organs	Heart

Q9. Neuron diagram

Answer (write in exam):

Neuron consists of:

- Cell body
 - Dendrites
 - Axon
-

Q10. Name the following

Answer:

- (a) Inner lining of mouth → **Epithelial tissue**
 - (b) Connects muscle to bone → **Tendon**
 - (c) Transports food → **Phloem**
 - (d) Stores fat → **Adipose tissue**
 - (e) Fluid connective tissue → **Blood**
 - (f) Brain tissue → **Nervous tissue**
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Q11. Identify tissue

Answer:

- Skin → Epithelial tissue
 - Bark → Protective tissue
 - Bone → Connective tissue
 - Kidney tubule → Epithelial tissue
 - Vascular bundle → Complex tissue
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Q12. Where is parenchyma found?

Answer:

- Cortex
 - Pith
 - Leaves
-

Q13. Role of epidermis in plants

Answer:

- Protects plant
 - Reduces water loss
 - Helps in gas exchange
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Q14. Cork as protective tissue

Answer:

- Cells are dead and compact
 - Prevent water loss
 - Protect against injury
-

Extra Questions

Q15. Tissue responsible for movement

Answer:

Muscular tissue

Q16. Structure of neuron

Answer:

Long cell with dendrites and axon.

Q17. Features of cardiac muscles

Answer:

- Involuntary
 - Striated
 - Rhythmic contraction
-

Q18. Functions of areolar tissue

Answer:

- Supports organs

- Fills spaces
 - Helps in repair
-

Q19. Types of simple tissues

Answer:

- Parenchyma
 - Collenchyma
 - Sclerenchyma
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Q20. Apical meristem location

Answer:

At tips of roots and shoots.

Q21. Husk of coconut

Answer:

Sclerenchyma

Q22. Constituents of phloem

Answer:

- Sieve tubes
 - Companion cells
 - Phloem parenchyma
 - Phloem fibres
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Q23. Utility of tissues

Answer:

- Division of labour
- Efficient functioning
- Specialisation

Q24. Are plant and animal tissues same?

Answer:

No, they are different because plants and animals have different functions and structures.

Key Points for Revision

- Tissue = group of similar cells
 - Xylem → water transport
 - Phloem → food transport
 - Muscles → movement
 - Neurons → control system
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Class 9 biology– Improvement in Food Resources (NCERT Solutions)

Q1. Explain any one method of crop production for high yield

Answer:

Use of High Yielding Variety (HYV) seeds

- HYV seeds produce more crops per unit area
 - Resistant to diseases
 - Require proper irrigation and fertilizers
 - Increase overall productivity
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Q2. Why are manure and fertilizers used?

Answer:

- Provide essential nutrients to crops
 - Improve soil fertility
 - Increase crop yield
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Q3. Advantages of intercropping and crop rotation

Answer:

Intercropping:

- Better use of nutrients
- Reduces pest attack
- Increases yield

Crop rotation:

- Maintains soil fertility
 - Prevents diseases
 - Improves productivity
-

Q4. What is genetic manipulation?

Answer:

Genetic manipulation is altering genes to improve crops.

Use:

- Disease resistance
 - Higher yield
 - Better quality crops
-

Q5. How do storage grain losses occur?

Answer:

- Due to pests (insects, rodents)
 - Moisture
 - Fungi and bacteria
 - Improper storage conditions
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Q6. Benefits of good animal husbandry

Answer:

- Increases milk, meat, egg production
 - Improves quality of products
 - Better income for farmers
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Q7. Desirable characters of bees

Answer:

- High honey collection capacity
 - Gentle nature
 - Low tendency to swarm
 - Disease resistant
-

Q8. What is pasturage?

Answer:

Availability of flowers for bees.

Relation:

- More flowers → more nectar → more honey
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Q9. How are fish obtained?

Answer:

- Capture fishing (natural water)
 - Fish farming (aquaculture)
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Q10. Advantages of composite fish culture

Answer:

- Different species use different food levels
 - No competition
 - Higher fish production
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Q11. Poultry efficiency statement

Answer:

Poultry converts low-quality feed into **high-protein food (eggs, meat)** efficiently.

Q12. Common practices in dairy and poultry

Answer:

- Proper feeding
 - Clean shelter
 - Disease control
 - Breeding management
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Q13. Broilers vs Layers

Answer:

Feature	Broilers	Layers
Purpose	Meat	Eggs
Growth	Fast	Moderate
Feed	High protein	Balanced diet

Q14. Method for improving cattle breeds

Answer:

Cross-breeding

Reason:

- Combines good qualities
 - Improves milk production
-

Q15. Why prefer preventive and biological methods?

Answer:

- Eco-friendly
 - Safe for humans
 - Prevents pollution
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Q16. Factors causing grain loss

Answer:

- Moisture
 - Insects
 - Rodents
 - Microorganisms
-

Q17. Best farming condition

Answer:

(c) Farmers use quality seeds + irrigation + fertilizers + protection

Reason:

Gives maximum yield and protection.

Q18. Manure vs Fertilizers**Answer:**

Manure	Fertilizers
Organic	Chemical
Improves soil structure	Quick nutrients
Eco-friendly	May cause pollution

Q19. Benefits of cattle farming**Answer:**

- Milk production
 - Agricultural work
 - Manure for fields
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Q20. Common factor in poultry, fisheries, bee-keeping**Answer:**

Improved **breeding and management practices**

Q21. Difference: fishing methods**Answer:**

Method	Meaning
Capture fishing	Natural water bodies
Mariculture	Sea water farming

Method	Meaning
Aquaculture	Controlled water farming

Key Points for Revision

- HYV seeds increase yield
 - Crop rotation maintains soil
 - Animal husbandry improves income
 - Fisheries + poultry = high protein food
 - Storage losses must be controlled
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